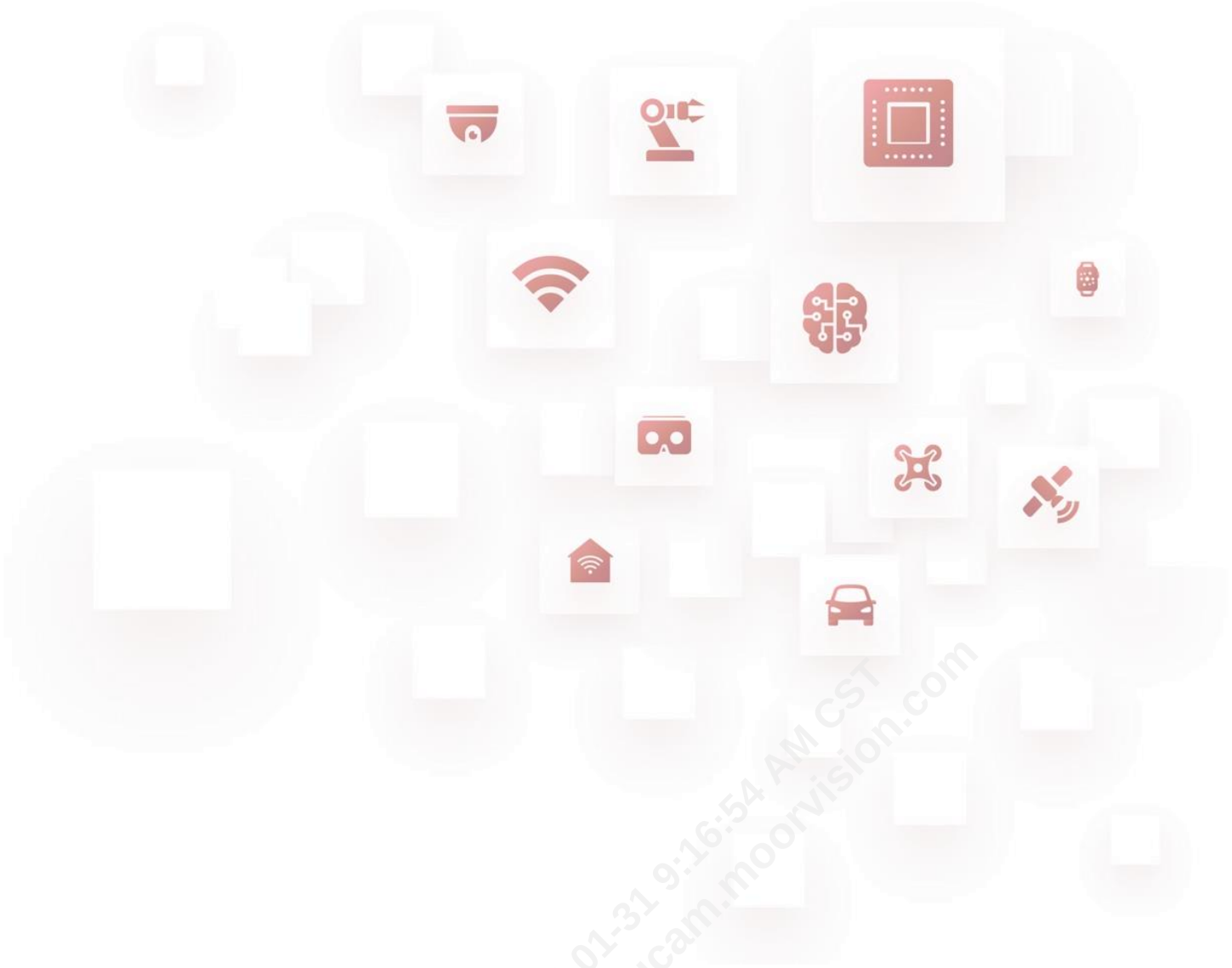
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AIOT ENABLER



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Thundercomm TurboX C410/C610 Open Kit Linux Embedded

Software Release Notes

Rev V1.0
June 16, 2023

DN: tc-gle-2410
SDK Version: LE2.0.Post-CS1.r002002

Revision History

Revision	Date	Author	Description
1.0	June 16, 2023	Xiaojun, Wang	Initial release.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Introduction

This document provides information on detailed steps to obtain, build, and program QCS610 LE (Linux Embedded) SDK (Software Development Kit) into the TurboX C410/C610 open kit.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark TURBOX.

The LE SDK supports the Linux operating system on the Thundercomm TurboX SOM. The LE SDK aims to provide a simple and flexible development platform to rapidly build IoT (Internet of Things) products with the QTI chipsets. The LE SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

The LE SDK is based on the Yocto Project 3.1 release and follows several of its development principles. The Yocto Project is used by the LE SDK to build framework.

Table 1-1. TurboX C410/C610 platform specifications

SOC	SKUx		Standard SKU[1)]	Thundercomm Product No. (TPN)	Model
QCS610	SKU1	2GB+16GB	Yes	C610-D21-369PWXX C610-D21-LCXPW00 (LE Specified)	TurboX C610
QCS610	SKU2	2GB+64GB	No	C610-D26-369PWXX C610-D26-LCXPW00 (LE Specified)	TurboX C610
QCS610	SKU3	4GB+64GB	Yes	C610-D46-369PWXX C610-D46-LCXPW00 (LE Specified)	TurboX C610
QCS410	SKU4	2GB+16GB	Yes	C410-D21-369PWXX C410-D21-LCXPW00 (LE Specified)	TurboX C410
QCS410	SKU5	2GB+64GB	No	C410-D26-369PWXX C410-D26-LCXPW00 (LE Specified)	TurboX C410
QCS410	SKU7	4GB+64GB	No	C410-D46-369PWXX C410-D46-LCXPW00(LE Specified)	TurboX C410
QCS410	SKU8	2GB+8GB	No	C410-D28-369PWXX C410-D28-LCXPW00 (LE Specified)	TurboX C410
QCS410	SKU9	4GB+16GB	No	C410-D41-369PWXX C410-D41-LCXPW00(LE Specified)	TurboX C410

➲ **NOTE:**

[1)] indicates standard SKU series. For standard SKU series, retail stock is available; as to non-standard SKU series, in case of any customer requirement, specific orders need to be placed accordingly.

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Chapter 2. Development Environment Setup

Refer to *TurboX SOM SDK Manager User Guide* to set up the development environment.

➲ **NOTE:** Make sure that you use the sdkmanager built based on Ubuntu 18.04. When you start the sdkmanager, the following window will appear.



Figure 2-1.sdkmanager Window

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Chapter 3. Get Started with LE SDK

3.1. LE SDK information

Table 3-1. LE SDK information

SDK Version	Baseline	Baseline Distribution	OS Version
LE2.0.Post-CS1.r002002	QCS610.LE.2.0	Post-CS[3]0.0.009.1	Linux Embedded 2.0

3.2. Block diagram

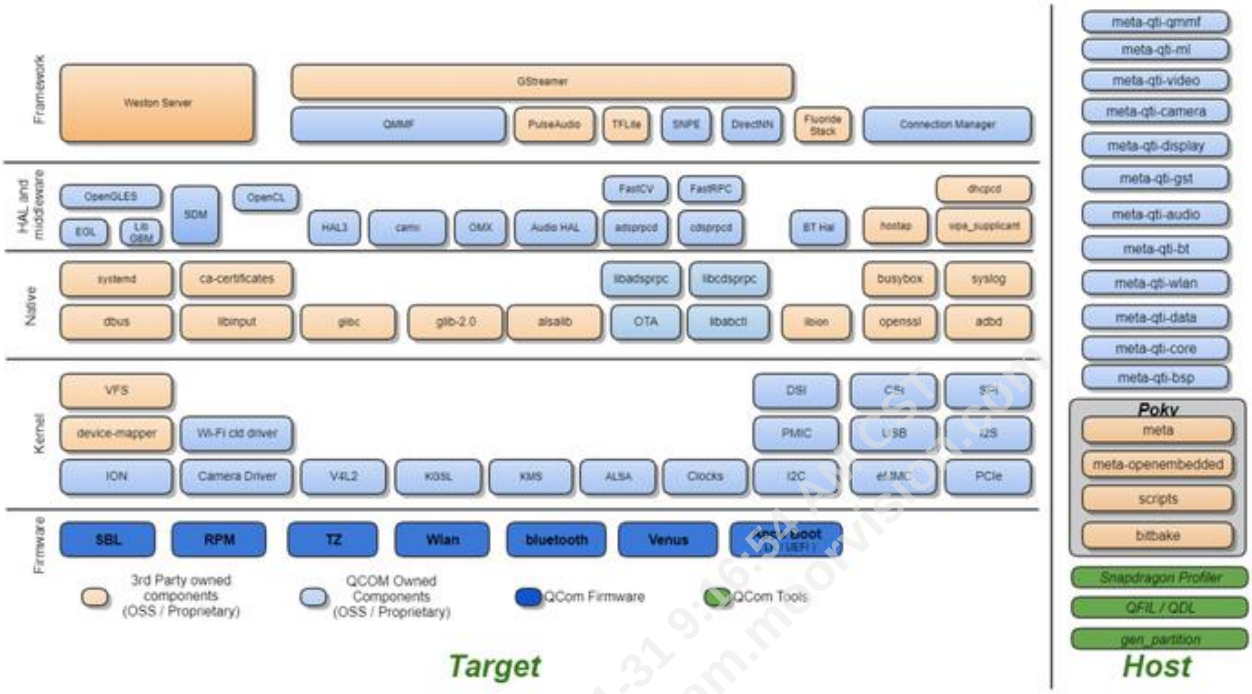


Figure 3-1.Block Diagram

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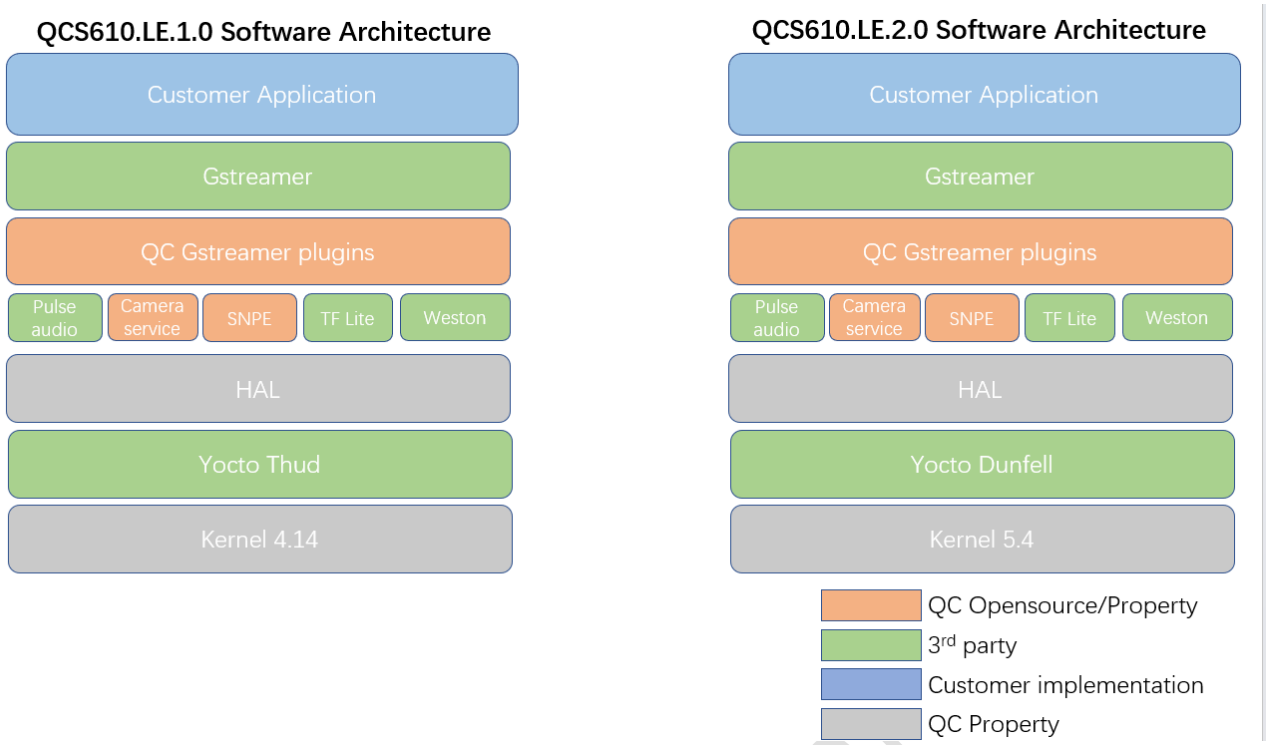


Figure 3-2.Difference Between QCS610 LE1.0 & QCS610 LE2.0

3.3. SDK source tree structure

```
SDK.Turbox_C610.LE2.0_CS/
├── about.html -> common/about.html      //SDK version info(Product,Distribution, …)
├── adsp_proc      //aDSP image is released as a binary and no build compilation is needed
├── aop_proc       //AOP image is released as a binary and no build compilation is needed
├── apps_proc
   ├── build-qti-distro-fullstack-virtualization-debug   /*Directory of temporary file and
image files generated by compilation */
   ├── disregard
   ├── downloads                    //Software packages required for compilation
   ├── poky                         //Yocto's recipe directory
   ├── prebuilt_HY11                //Qcom privatized software package
   ├── setup-environment -> poky/qti-conf/set_bb_env.sh     //Script for set the build
environement
   ├── src        //HLOS source code, such as linux kernel, filesystem, vendor, OTA, etc.
   ├── sstate-cache
   └── syncbuild.sh
├── boot_images  //BOOT image is released as a binary and no build compilation is needed
├── BP-CODE        //non-HLOS binary files
├── btfm_proc      //BTFM image is released as a binary and no build compilation is needed
├── cdsp_proc      //cDSP image is released as a binary and no build compilation is needed
├── common/config/emmc/partition.xml          //eMMC partition file
├── contents.xml -> common/contents.xml       //image contents file
├── modem_proc    //MODEM image is released as a binary and no build compilation is needed
├── trustzone_images //TZ image is released as a binary and no build compilation is
needed
├── turbox
│   └── tools
│       ├── build_script                      //SDK build scripts
│       └── flash_tools                       //Flash tools files(qdl, …)
├── turbox_build.sh -> turbox/tools/build_script/common/turbox_build.sh //SDK build
script
├── turbox_flash.sh -> turbox/tools/build_script/device/turbox_flash.sh //Fastboot flash
script
├── venus_proc   //VIDEO image is released as a binary and no build compilation is needed
├── wdsp_proc    //wDSP image is released as a binary and no build compilation is needed
└── wlan_proc    //WLAN image is released as a binary and no build compilation is needed
```

3.4. Download LE SDK

Refer to sections "*Initialize sdkmanager APP*" and "*Download SDK*" of *TurboX SOM SDK Manager User Guide* to set up the development environment and download SDK source code.

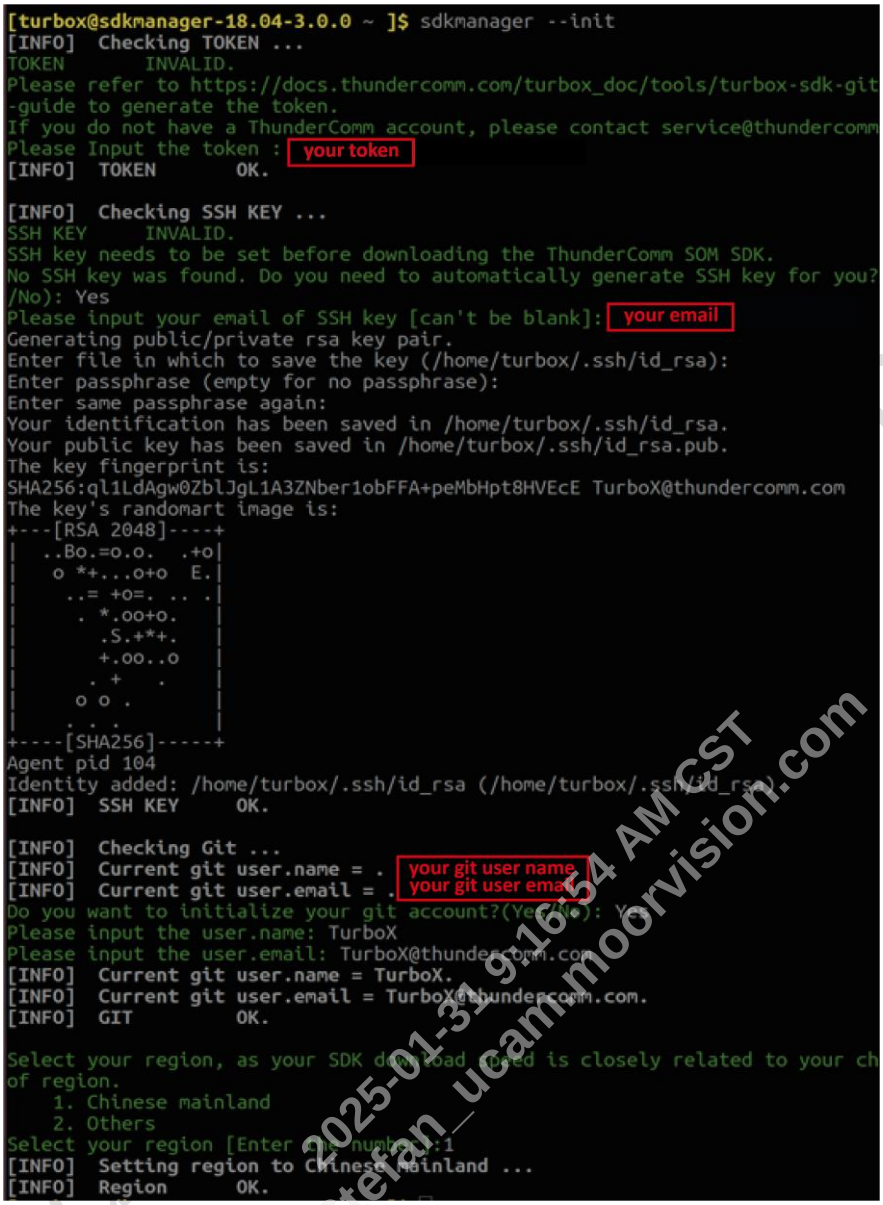


Figure 3-3.Initialize sdkmanager APP

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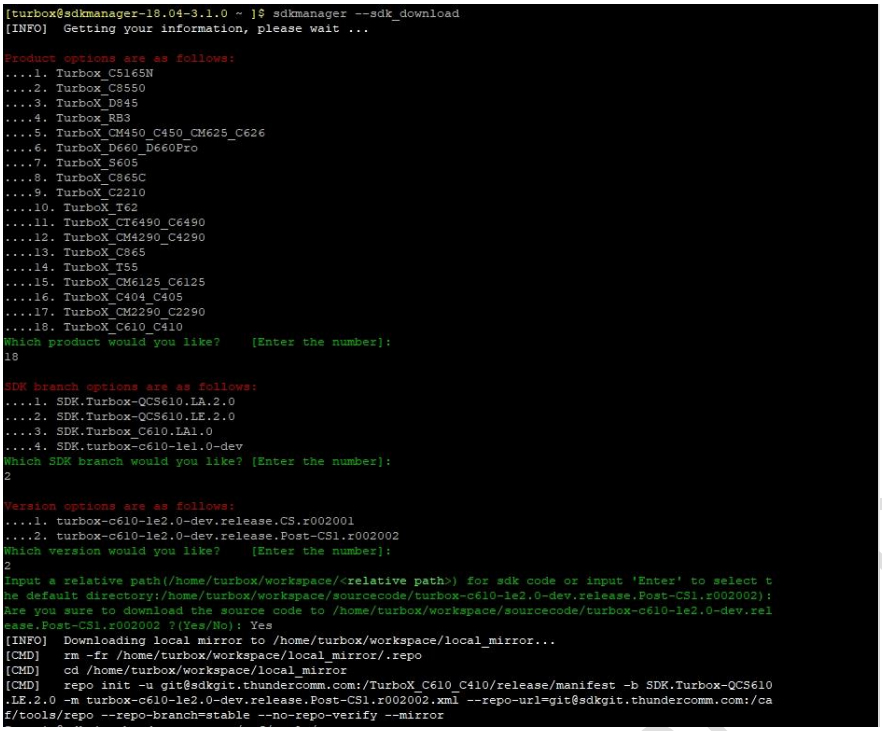


Figure 3-4.Download SDK

3.5. Build image

To compile the HLOS, run the following script file according to your image version in the root directory.

➲ **NOTE:** Please install the necessary compilation tools:

```
turbox@sdkmanager-18.04-3.0.0:~ $sudo apt-get install uuid-dev
```

- Build debug version:

```
turbox@sdkmanager-18.04-3.0.0:~ $ ./turbox_build.sh -a -l -v debug
```

- Build perf version:

```
turbox@sdkmanager-18.04-3.0.0:~ $ ./turbox_build.sh -a -l -v perf
```

➲ **NOTE**: UART debug cannot be used in perf version.

- Run the following commands to rebuild *kernel* additionally (if necessary).

```
$ cd apps_proc
$ MACHINE=qcs610-odk-64 DISTRO=qti-distro-fullstack-debug
$ source poky/qti-conf/set_bb_env.sh
$ bitbake -fc do_makeboot qti-multimedia-image
```

➲ **NOTE**: The default *boot.img* has verify enabled and cannot be flashed unless the kernel is rebuilt. Hence, when rebuilding only the kernel, you need to flash the *boot.img.noverity* which has verify disabled, refer to *turbox_flash.sh* to flash the *boot.img.noverity*.

- Run the following commands to rebuild *dtbo* additionally (if necessary).

```
$ cd apps_proc
$ MACHINE=qcs610-odk-64 DISTRO=qti-distro-fullstack-debug
$ source poky/qti-conf/set_bb_env.sh
$ bitbake -c cleanall linux-msm && bitbake linux-msm
```

3.6. Pack ROM package

Step 1. Run the following command to update common information according to your image version.

- **Debug version:**

```
turbox@sdkmanager-18.04-3.0.0:~ $  ./turbox_build.sh -u --uci-imf -v debug
```

- **Perf version:**

```
turbox@sdkmanager-18.04-3.0.0:~ $ ./turbox_build.sh -u --uci-imf -v perf
```

➲ **NOTE:** The command is successfully executed with the following log displayed.

```
- update_common_info.py:======  UPDATE COMMON INFO COMPLETE=======
```

Step 2. Run the following command to pack the ROM package. You can find the packed ROM package at *turbox/output*. Replace `<debug/perf>` with `debug` or `perf`.

- Pack Fastboot image

```
turbox@sdkmanager-18.04-3.0.0:~ $ ./turbox_build.sh --zip_fastboot_build -l -v
<debug/perf>
```

- Pack FlatBuild image

```
turbox@sdkmanager-18.04-3.0.0:~ $ ./turbox_build.sh --zip_flat_build -l -v
<debug/perf>
```

➲ **NOTE:** Customers can also get the ROM package provided by Thundercomm from:

- **Debug version:**
  - *FlatBuild_TurboX_C610_xx.xx_LE2.0.R.debug.Post-CS1.r002002.zip*

    md5: 26db89c3131ef0dc9b40940febb8fa40
  - *FastbootBuild_TurboX_C610_xx.xx_LE2.0.R.debug.Post-CS1.r002002.zip*

    md5: 27c049849a210df98e78bdefb89ad5db
- **Perf version:**
  - *FlatBuild_TurboX_C610_xx.xx_LE2.0.R.perf.Post-CS1.r002002.zip*

    md5: f1ca69ba782b664b76128be87a2387a8
  - *FastbootBuild_TurboX_C610_xx.xx_LE2.0.R.perf.Post-CS1.r002002.zip*

    md5: 6fea4fb2dd70f283ae908943f2affff7
- **OTA full package:**

  update_ext4_Turbox_C610_xx.xx_LE2.0.R.debug.Post-CS1.r002002.zip

  md5: eadbc2c0c60e91fa4b9ff023140566ea
- **Base image resource package:**

  target-files-ext4_Turbox_C610_xx.xx_LE2.0.R.debug.Post-CS1.r002002.zip

  md5: 77d8aaa0018d308f7d6970d9b3dfb9ba

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3.7. Flash images to board

➲ **NOTE**: If the device fails to boot up after image flash with **Fastboot**, try to flash the **FlatBuild** image again with *turbox_flash_flat.sh or TurboX Easy Flash*.

3.7.1. Flash images on Windows

You can flash images on Windows by using TurboX Easy Flash, which is a tool created by Thundercomm to help SOM developers easily flash device image, backup and restore data, etc.

For tool acquisition and operation procedures, refer to *TurboX Easy Flash User Guide*.

3.7.2. Flash images on Linux

- **Flash images on Linux with Fastboot**

  Step 1. Enter the **Fastboot** mode.

  Option 1. Enter the following commands when the device is in **ADB** mode.

```
$ adb root
$ adb reboot bootloader
```

  Option 2. Power off the device, hold the **Volume down** button and click the **Power on** button, then release the **Volume down** button about 5 seconds later.

  ➲ **NOTE**: To check if the device has entered the **Fastboot** mode, run the following command:

```
$ sudo fastboot devices
```

  The process is successful with the following output displayed on the shell.

```
21f10525  fastboot
```

  Step 2. Flash images.

  Execute the commands below to flash images to the board. Replace `<folder name of image package>` with the actual image package name of your product.

```
$ cd <folder name of image package>
$ sudo ./turbox_flash.sh --all -r -f -v <ROM type>
```

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➲ **NOTES:**

- Please replace **<ROM type>** with **<debug>** (for debug version) or **<perf>** (for perf version) as required. Do not enter the angle brackets "**<>**" which are used for indicating code variables.
- Enter **./turbox_flash.sh -h** to acquire detailed usage.

```
$ ./turbox_flash.sh -h
Usage:
  bash ./turbox_flash.sh [options] [options]
  Images except gpt,userdata,persist will be flashed if no options input

Description:
  flash images with fasboot

Options:
  -h, --help   display this help message
  -v, --variant <debug/perf/user>    set the build variant.(default: debug)
  -f, --force  forcibly flash gpt,persist and userdata without confirmation.
             User data and calibration data(BT/WIFI MAC address, etc..) will
be ereased.
  -r, --reboot     reboot the device automatically after flashing completes.
  -b, --nonhlos    flash all bp images(xbl,aop,ftfm,tz,dsp,modem,pmic).
  -a, --hlos       flash all ap
images(aboot,bootimage,recovery,system,vendor,cache,mdtp,userdata,persist).
  --all       flash all images

Example:
  ./turbox_flash.sh
     Flash the images except gpt,userdata,persist.

  ./turbox_flash.sh --force
     Flash all images.

  ./turbox_flash.sh –hlos
Flash all ap images.
```

- **Flash images on sdkmanager with *turbox_flash_flat.sh* tool**

  ➲ **NOTE:** Please update the sdkmanager version to v1.2.0 or latter which has built-in *turbox_flash_flat.sh*.

  Step 1.Stop the ModemManager on your host PC and close the application that occupies the ttyUSB0 port, such as **Minicom**.

  ```
  $ sudo systemctl stop ModemManager
  ```

  Step 2.Prepare the FlatBuild images.

  Check if the images are ready and exists in sdkmanager, then enter the image folder. Replace the image package name with the actual image package name of your product.

  ```
  turbox@sdkmanager-18.04-3.0.0:~/workspace $ cd <folder name of image package>
  ```

  Step 3.Enter **9008 Mode**.

  1) Power off the device by disconnecting the power cable and USB cable.
  2) Connect the power cable.
  3) Hold the **Force USB Boot** button.
  4) Press the **Power on** button and wait for about 3 seconds.
  5) Release the **Force USB Boot** button and **Power on** button.
  6) Connect the Type-C USB cable to an Ubuntu computer.

  To check if your device has entered the **9008 Mode**, run the following command:

  ```
  turbox@sdkmanager-18.04-3.0.0:~ $ lsusb
  ……
  Bus 001 Device 042: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
  …… .
  ```

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Step 4.Start to flash the FlatBuild images.

Run the following command under the image folder.

```
turbox@sdkmanager-18.04-3.0.0:~/workspace/<folder name of image package> $ turbox-flash-flat.sh
```

If the flashing process is successful, the following log will be printed.

```
11:20:09: INFO:     _           (done)
11:20:09: INFO:    | |
11:20:09: INFO:  __| | ___  _ __   ___
11:20:09: INFO:  / _` |/ _ \| '_ \ / _ \
11:20:09: INFO: | (_| | (_) | | | |  __/
11:20:09: INFO:  \__,_|\___/|_| |_|\___|
11:20:09: INFO: {All Finished Successfully}

11:20:09: INFO: Overall to target 30.119 seconds (85.78 MBps)
11:20:09: INFO: {percent files transferred 100.00%}
Writing log to '/home/turbox/workspace/FlatBuild_Turbox-C610_xx.xx_le2.0.D.debug.20221027.2351/port_trace.txt', might take a minute
```

The device board will automatically reboot.

Step 5.Connect the USB 3.1 Gen 1 Type-C cable and check the ADB device with the following command.

```
$ adb devices
List of devices attached
ac55dc38      device
```

➲ **NOTE:** For more information, please refer to: *Turbox_flash_flat-user-guide*.

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Chapter 4. Features

This chapter introduces the software features supported by the DK.

4.1. Presented features

This section describes the key software features that have been included in earlier versions.

- **SDK**
  - OpenEmbedded Linux 2.0
  - Yocto Dunfell(v3.1)
  - GCC toolchain (v8.2.0)
  - glibc (v2.28)
  - systemd (v1.239)
- **Platform**
  - Linux Kernel 5.4
  - 64-bit only
  - UBWC
  - OTA
  - HY22
  - DTB Overlay
- **CV**
  - FastCV
- **Security**
  - DM-Verity
  - SELinux
  - KASan
  - FDE
  - VBLE
  - Non-Root Processes
  - SecureBoot
- **Connectivity**
  - WLAN 802.11ac
  - USB 3.1 Gen 1
  - Bluetooth 5.0
  - WPA-R3
- **Storage**
  - eMMC
  - UFS 3.1

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- **Multimedia**
  - Weston 8.0
  - GStreamer (v1.16) Framework
  - Hexagon-SDK
- **Camera**
  - Staggered HDR (v2&v3.8)
  - EIS (v2.0)
  - IRLED
  - LDC and EIS
  - Dual Independent Cameras
  - Force sensor mode to bypass default logic
  - Manual disablement of IR cut filter with IR LED OFF
  - Multiple black level
- **Video**
  - JPEG DMA Engine
  - H264 & HEVC encode and decode
  - UVC & UAC
- **Audio**
  - AAC, PCM and MP3 (only playback) Encode and Playback
  - Audio + Hands Free Profile concurrency
  - Voice UI 3.0
- **Display**
  - DP
  - LCD
- **Graphics**
  - OpenGL ES 3.2
  - OpenCL 2.1
- **AI**
  - BENCHMARK APP
  - AI Inferencing via SNPE v1.36
  - AI Inferencing via TFLite 2.0
  - TFLite supports NNAPI delegate offloading graph processing to DSP
  - SNPE run on GPU CPU and DSP
  - TF Lite DSP run-time support through HexNN
- **Battery**
  - Software-based fuel gauge solution for battery powered devices

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4.2. New features

This section describes the new key software features included in this SDK version.

- Support HDMI-IN
- Enable BT/Wi-Fi/59/Red/Green/Blue LED
- Enable LSM6DSOTR A+G sensor
- Enable DP-HDMI adapter
- OTA

4.3. On-going features

This section describes the software features that will be included in future SDK versions.

- UVC&UAC device

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Chapter 5. Test Report

Table 5-1. Test report

Test Case	Result
FlatBuild version flash	Pass
Fastboot version flash	Pass
Metabuild version flash	Pass
Boot_adb reboot	Pass
Boot_adb reboot bootloader	Pass
Boot_fastboot reboot	Pass
Fastboot (adb & volume down)	Pass
Power on button	Pass
Volume up/down button	Pass
Force USB Boot button	Pass
Memory Information	Pass
64-bit user space	Pass
Disable verity	Pass
SD Card auto mount/unmount/multiple plug & unplug	Pass
SD card file push/pull	Pass
Multimedia_GST_recorder_4k_30_AVC_MP4_mux	Pass
Multimedia_GST_recorder_4k_30_HEVC_MP4_mux	Pass
Multimedia_GST_recorder_1080p_30_AVC_filesave	Pass
Multimedia_GST_Snapshot_One 1080p AVC mp4 muxed video with JPEG snapshots	Pass
Multimedia _GST_Overlay_1080p_30 YUV with UserText overlay	Pass
Multimedia _GST_Display_1080p_30 Live preview on a local display	Pass
Multimedia _GST_Display_1080p_30 Live preview over TCP/RTSP	Pass
Camera_LDC/ZOOM	Pass
Camera_Dual Independent Cameras (IMX577)	Pass
Audio Line IN	Pass
Audio_1-6 D-MIC recorder individually	Pass
Audio_1-6 channels recorder concurrency with GST	Pass
Audio_1-6 channels recorder concurrency with parecord	Pass
Audio headphone	Pass
Audio speaker	Pass
Display_DP	Pass
Display_LCD	Pass

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Test Case	Result
Display_Weston color Bar	Pass
USB_Type-C functionality	Pass
USB_Type-A functionality	Pass
USB_Micro for UART	Pass
Ethernet_RJ45 connect	Pass
Ethernet_Ping test	Pass
Connectivity_Bluetooth	Pass
Connectivity_Wi-Fi_connect to AP	Pass
Battery	N/A
Charge the battery via Type-c	N/A
12V DC power supply	Pass
AI inferencing via TFLite 2.0	Pass

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Chapter 6. Known Issues

Table 6-1. List of known issues

Issue No.	Description	Status
Limitations-002	RJ45 indicator is abnormal.	Open
BUG-13155	System cannot stay Sleep Mode.	Ongoing
BUG-13253	Ethernet RGMII cannot work in 10M/s and 100M/s modes.	Ongoing
BUG-Q-3147869	[QCS610.LE.2.0][UEFI]: Charger test fails with ERR 3.	Closed
BUG-Q-3250714	[QCS610.LE.2.0][C610+IMX334]: “Segmentation fault” appears during the switchover between ready state and playing state in storage use case.	Closed

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